Probability, conditional probability, Bayes' theorem

Inference labs use the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Distinguish marginal, joint, and conditional probability and compute each from a contingency table.
- State Bayes' theorem and apply it to a medical-testing scenario.
- Simulate a population and recover conditional probabilities by counting.

Prerequisites

Basic probability.

Background

Probability is the language for reasoning under uncertainty. In biostatistics, most of the probabilities we care about are conditional — the probability that a patient has a disease *given* a test result; the probability of survival *given* a treatment; the probability of an adverse event *given* a risk factor. The distinction between marginal and conditional probability is the distinction between "how common is X in the population" and "how common is X in a particular subgroup".

Bayes' theorem is the rule for flipping a conditional probability. Given $P(\mathrm{A}|\mathrm{B})$ and the marginal probabilities $P(\mathrm{A})$ and $P(\mathrm{B})$, it returns $P(\mathrm{B}|\mathrm{A})$. In medical testing this lets us move from *how often does the test fire when the disease is present* (sensitivity) to *how likely is disease given a positive test* (positive predictive value). The two are routinely confused; Bayes' theorem is the cure.

The counterintuitive part of Bayes is the role of prevalence. For a test with fixed sensitivity and specificity, PPV rises and falls with the prevalence in the tested population. Screening a very-low-risk population with an imperfect test guarantees that most positives will be false positives, regardless of the test's quality.

Setup

```
library(tidyverse)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

We are not testing a hypothesis in the Fisherian sense; we are computing a conditional probability. The quantity of interest is the positive predictive value of a test with known sensitivity and specificity in a population of known prevalence.

2. Visualise

Simulate a population of 100,000 people. Prevalence of disease = 1%. Test sensitivity = 95%. Test specificity = 95%.

```r
prev <- 0.01
sens <- 0.95
spec <- 0.95

pop <- tibble(
  id = seq_len(N),
  disease = rbinom(N, 1, prev)
) |>
  mutate(
    test = if_else(disease == 1,
                   rbinom(N, 1, sens),
                   rbinom(N, 1, 1 - spec))
  )
```

```r
tab
```

```r
as_tibble(tab) |>
  mutate(disease = recode(disease, `0` = "no disease", `1` = "disease"),
         test    = recode(test,    `0` = "neg", `1` = "pos")) |>
  ggplot(aes(test, disease, fill = n)) +
  geom_tile() +
  geom_text(aes(label = n), colour = "white") +
  scale_fill_viridis_c() +
  labs(x = "Test result", y = "True state", fill = "Count")
```

3. Assumptions

We assume each person is independent, the test's operating characteristics are fixed, and no selection bias. Real screening has verification bias, spectrum bias, and imperfect gold standards; we are ignoring all of them to isolate the Bayesian calculation.

```r
p_positive <- mean(pop$test)
p_pos_given_disease <- mean(pop$test[pop$disease == 1])
p_pos_given_no_disease <- mean(pop$test[pop$disease == 0])

tibble(quantity = c("P(D)", "P(T+)",
                    "P(T+|D)", "P(T+|~D)"),
       value = c(p_disease, p_positive,
                 p_pos_given_disease, p_pos_given_no_disease))
```

4. Conduct

Apply Bayes' theorem by hand.

```r
ppv_bayes

# By direct counting from the simulation.
ppv_count <- mean(pop$disease[pop$test == 1])
ppv_count
```

Same quantity, two routes. Bayes is algebra on the marginals; simulation is counting from the table. They must agree, within sampling error.

```r
sweep <- tibble(
  prev = seq(0.001, 0.5, length.out = 200),
  ppv = (sens * prev) / (sens * prev + (1 - spec) * (1 - prev))
```

```
)
```

```
sweep |>
  ggplot(aes(prev, ppv)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 0.5, linetype = 2, colour = "grey50") +
  labs(x = "Prevalence", y = "Positive predictive value")
```

5. Concluding statement

> With sensitivity = `sens` and specificity = `spec`, a test applied in a population of prevalence = `prev` has a positive predictive value of `round(ppv_bayes, 3)` — that is, fewer than one in five positive results is a true positive. In a simulated population of `format(N, big.mark = ",")`, the PPV recovered by direct counting was `round(ppv_count, 3)`. Negative predictive value was correspondingly high.

When prevalence is low, even a very accurate test produces mostly false positives. This is why population screening decisions are controversial, and why "my test is 95% accurate" is an incomplete statement.

Common pitfalls

- Confusing sensitivity with PPV. $P(\mathrm{T+}|\mathrm{D})$ and $P(\mathrm{D}|\mathrm{T+})$ are not the same.
- Reporting a test's "accuracy" without specifying prevalence.
- Assuming independence of repeated tests without checking.
- Treating the prior (prevalence) as negotiable evidence.

Further reading

- Gigerenzer G. *Reckoning with Risk*.
- Altman DG & Bland JM (1994), *Diagnostic tests 2: predictive values*.

Session info

```
```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)